TECHNICAL MANUAL

BLAND2GRAND

Maker's Guide

The Smart, Autonomous Carousel Spice Dispenser

VERSION 1.0 | REV. July 30, 2026 | DOC. B2G-MG-001

Contents

1 Introduction

This Maker's Guide is written for someone who has the printed/purchased parts to build one in front of them and wants to assemble it and get it running. It covers two supported controller boards:

Arduino UNO Q
The primary, currently supported build. The Flask backend, dispense logic, and MCU sketch all run on a single board, launched through Arduino App Lab.

Arduino UNO R4 WiFi
The original build. The backend and frontend run on a separate laptop, communicating with the Arduino over a mobile hotspot.

Follow **Section 2** to physically assemble the machine, then **Section ??** to install and launch the software for whichever board you have.

At a Glance

Build Time	Skill Level	Boards Supported
One week for printing and assembly, and one afternoon for software	Intermediate: soldering, 3D printing, basic CLI	UNO Q (recommended), UNO R4 WiFi

2 Assembly

2.1 Printing the Parts

Printer and Material Notes

All printing for this build was done on a **Bambu P1S** and an **A1 mini** in **Bambu Studio**. Every part is printed in **PETG**. Unless a setting is called out below, use the slicer's default profile. If your slicer doesn't expose a listed setting, its own default is a safe substitute. **A 3D printer with a minimum build volume of 257 mm × 257 mm × 257 mm is required.**

Every part in this guide starts from Bambu Studio's stock **0.20mm Standard @BBL X1C** profile. Only the settings listed under each category below are changed; everything else is left at default.

Category Legend

Parts are grouped into three categories by print requirements. This legend is used consistently for the rest of this section:

Category	Profile Summary
🔴 **Category 1** Lightweight	1 wall loop, 10% infill/wall overlap, support on (tree)
🟠 **Category 2** Strong	3 wall loops, 30% infill/wall overlap, support on (tree)
🔵 **Category 3** Misc	Its own profile: Arachne walls, 3 wall loops, support on (normal)

Full Parts List

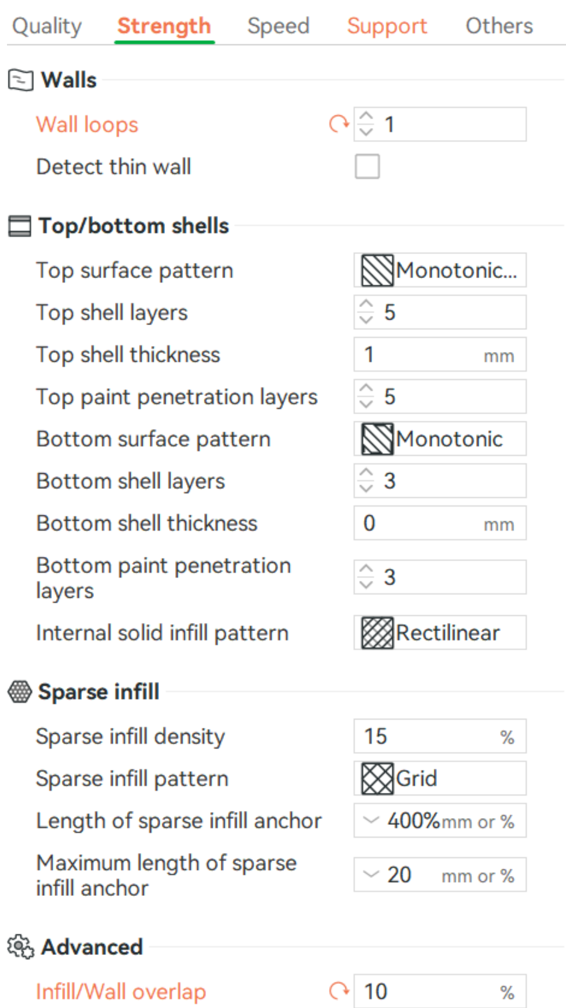
Part	Qty
Bevel Gear	8
Auger Power Shaft	8
Auger Power Gear (Full Spur)	8
Auger Motor Mount (Shaft Adapter)	1
Auger Motor Mount	1
Auger	8
Auger Bevel Attachment	8
Auger End Cap	8
Base BL, BR, FL, FR	1 each
Base Plate BL, BR, FL, FR	1 each
Carousel Drive	1
Carousel Motor Mount	1
Carousel Motor Mount (Spacer)	1
Carousel Track (Upper Trunk)	1
Carousel Track (Support Arm)	7
Funnel Support Extension	1
Internal Gear	1
Motor Shaft Auger Power Gear (Auger Half Spur)	1
Plate Holder	1
Segment	8
Segment (Lid)	8
Segment (Insert)	8
Segment (Fork)	8
Segment (Container)	8
Support Plate	1
Support Ring (Back Ring Track)	2

Part	Qty
Support Ring (LS Ring Track)	1
Support Ring (RS Ring Track)	1
Cycloidal Gearbox (Cover)	1
Cycloidal Gearbox (Disc)	1
Cycloidal Gearbox (Hover)	1
Cycloidal Gearbox (Input)	1
Cycloidal Gearbox (Output)	1

Category 1: Lightweight

Part	Qty
Base BL, BR, FL, FR	1 each
Base Plate BL, BR, FL, FR	1 each
Carousel Track (Upper Trunk)	1
Carousel Track (Support Arm)	7
Funnel Support Extension	1
Plate Holder	1
Segment	8
Segment (Lid)	8
Segment (Insert)	8
Segment (Fork)	8
Segment (Container)	8

Setting	Value
Wall Loops	1
Infill / Wall Overlap	10%
Support	On (Tree)



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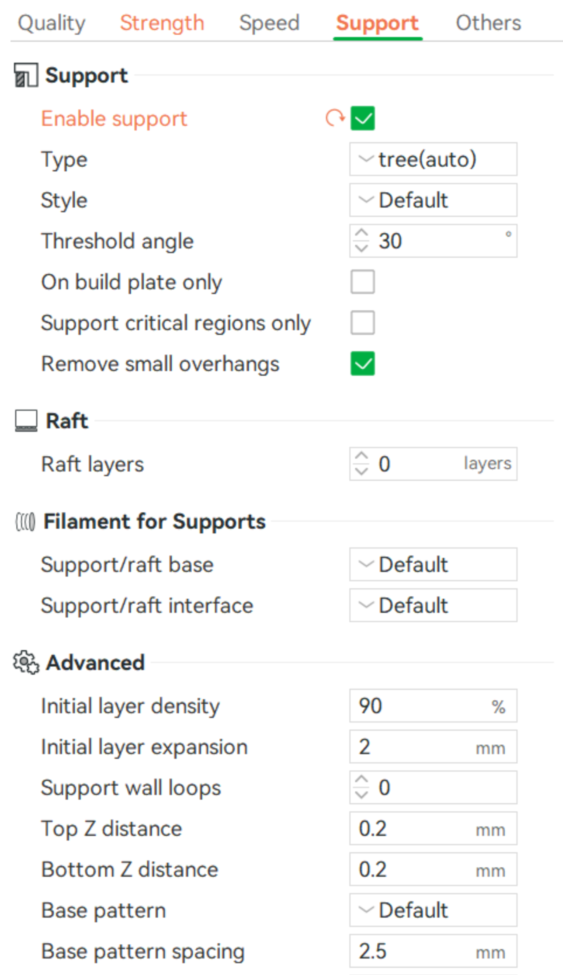


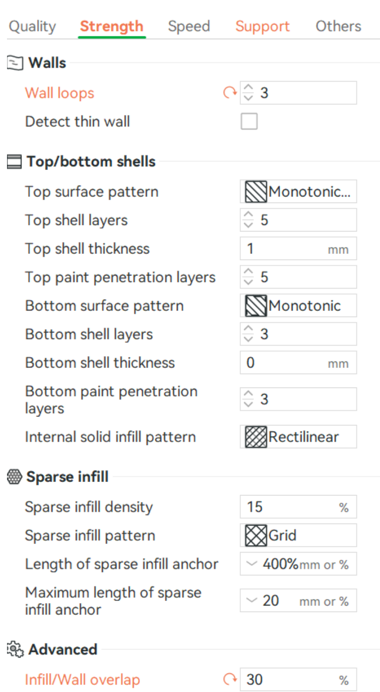
Support tab

Figure 1: Slicer settings for Category 1 (Lightweight) parts.

Category 2: Strong

Part	Qty
Carousel Drive	1
Carousel Motor Mount	1
Carousel Motor Mount (Spacer)	1
Internal Gear	1
Motor Shaft Auger Power Gear (Auger Half Spur)	1
Support Plate	1
Cycloidal Gearbox (Cover)	1
Cycloidal Gearbox (Disc)	1
Cycloidal Gearbox (Hover)	1
Cycloidal Gearbox (Input)	1
Cycloidal Gearbox (Output)	1

Setting	Value
Wall Loops	3
Infill / Wall Overlap	30%
Support	On (Tree)



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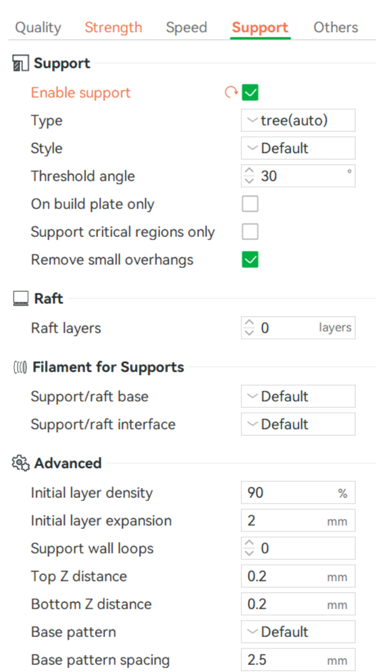


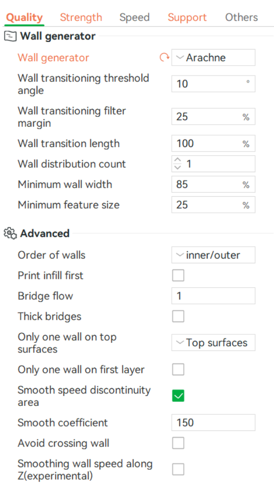
Support tab

Figure 2: Slicer settings for Category 2 (Strong) parts.

Category 3: Misc

Part	Qty
Auger	8

Setting	Value
Wall Generator	Arachne
Layer Height	0.2 mm (initial 0.16 mm)
Wall Loops	3
Support	On (Normal)



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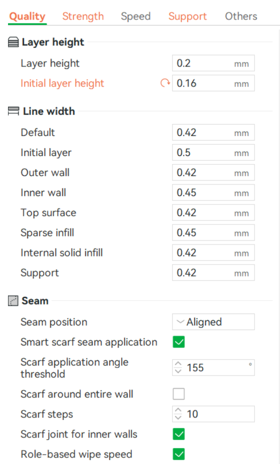

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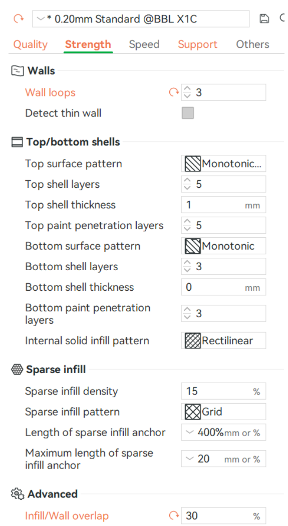

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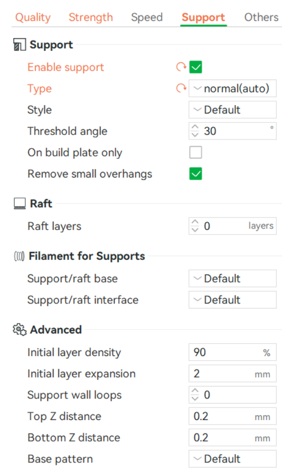


Support tab

Figure 3: Slicer settings for Category 3 (Misc) parts, the Auger.

2.2 Mechanical Assembly

This section walks through the full mechanical build of Bland2Grand in eight parts: the auger motor, the carousel motor and cycloidal gearbox, the base, the load cell, the carousel segments, the tree trunk, and final assembly. Build the sub-assemblies in order, as each part is set aside and used in a later step.

Before You Start

Have all printed parts from **Section 2.1** sorted and within reach, along with super glue (or threadlocker), a hex key set, and the hardware called out in each step.

Part 1: Auger Motor Assembly

1. With the Auger Motor Mount (Shaft Adapter) and the half spur gear, super glue the gear onto the shaft as shown below.

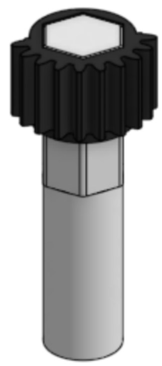


Figure 4: Half spur gear glued onto the auger motor shaft adapter.

2. Place the auger motor into the mount through the slit. Insert the four #6-32 $\frac{3}{8}$" bolts into the motor (the motor is tapped, so no nuts are needed on this side).

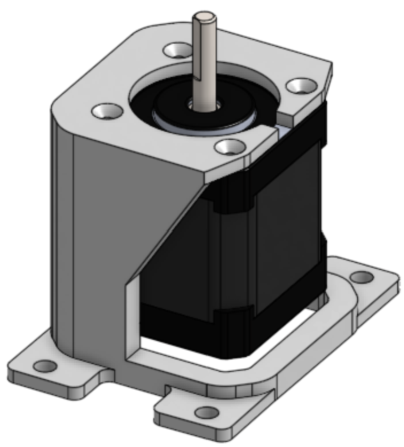


Figure 5: Auger motor seated into the Auger Motor Mount, viewed from the shaft side.

3 A second view of the same joint, showing the four #6-32 $\frac{3}{8}$″ bolts driven into the motor's tapped mounting holes.

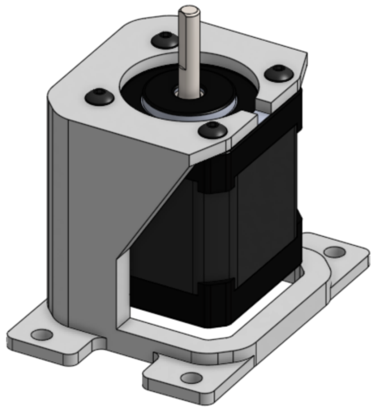


Figure 6: Auger motor mount, alternate angle showing #6-32 $\frac{3}{8}$″ bolt placement.

4 Slide the glued gear and shaft adapter (from **Step 1**) onto the motor shaft.

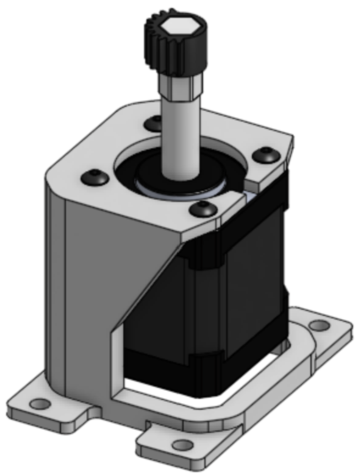


Figure 7: Gear and shaft adapter slid onto the auger motor shaft.

Set the completed Auger Motor assembly aside.

Part 2: Carousel Motor and Gearbox Assembly

5. Place the carousel motor into the Carousel Motor Mount, ensuring the holes on the mount line up with the holes on the motor.

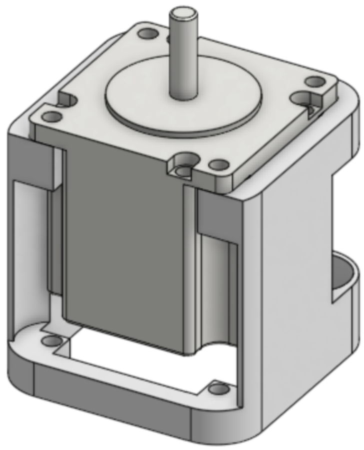


Figure 8: Carousel motor aligned with the Carousel Motor Mount.

6. Place the gearbox spacer and housing onto the motor, ensuring that all mounting holes are properly aligned.

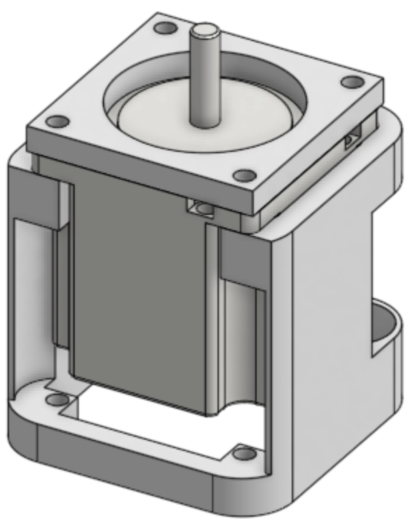


Figure 9: Gearbox spacer and housing installed onto the motor.

7 With the motor secured, the cycloidal gearbox housing sits on top, ready for the disc and input adapter.

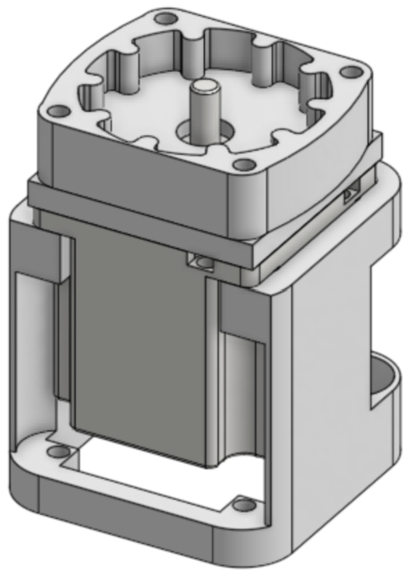


Figure 10: Cycloidal gearbox housing positioned above the mounted carousel motor.

8 Place the Cycloidal Gearbox (Disc) into the housing, then insert the input adapter onto the motor shaft and into the disc.

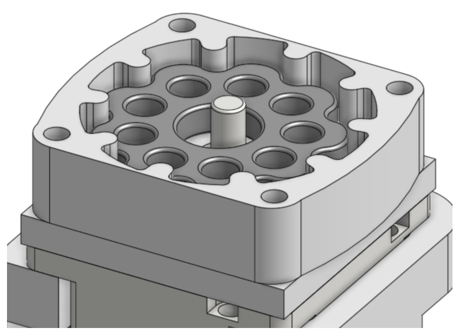


Figure 11: Cycloidal disc lowered into the housing with the input adapter engaging the motor shaft.

9 A closer view showing the input adapter fully seated through the disc's central bore.

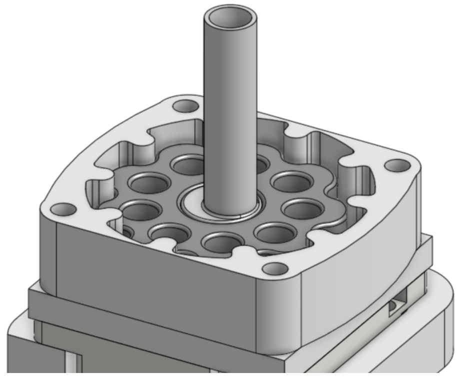


Figure 12: Input adapter seated through the cycloidal disc.

10 Insert the output carriage into the holes in the cycloidal disc, as shown below.

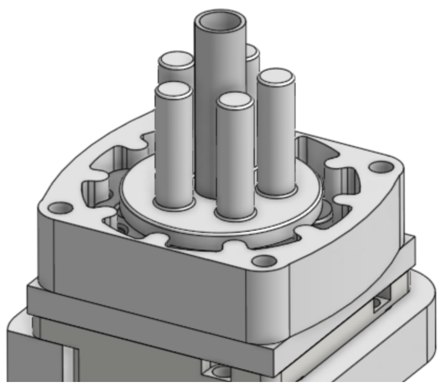


Figure 13: Output carriage pins inserted into the cycloidal disc's offset holes.

11 The Cycloidal Gearbox (Output) plate, which the carriage pins pass through to transmit rotation.

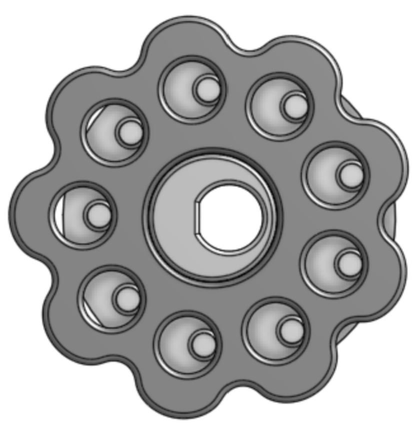


Figure 14: Cycloidal Gearbox (Output) plate.

12 Place the carousel drive gear in the cover between the top and bottom plates, as shown below. Then lower both parts onto the gearbox assembly.

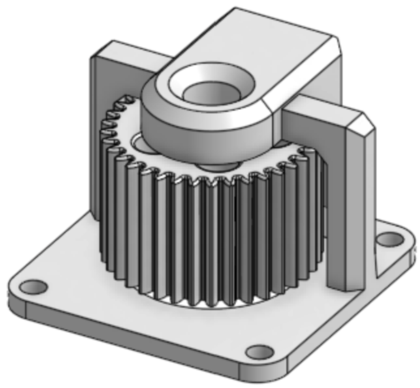


Figure 15: Carousel drive gear seated in its cover between the top and bottom plates.

13 An exploded view of the same joint, showing the drive gear, cover, and gearbox stack in assembly order.

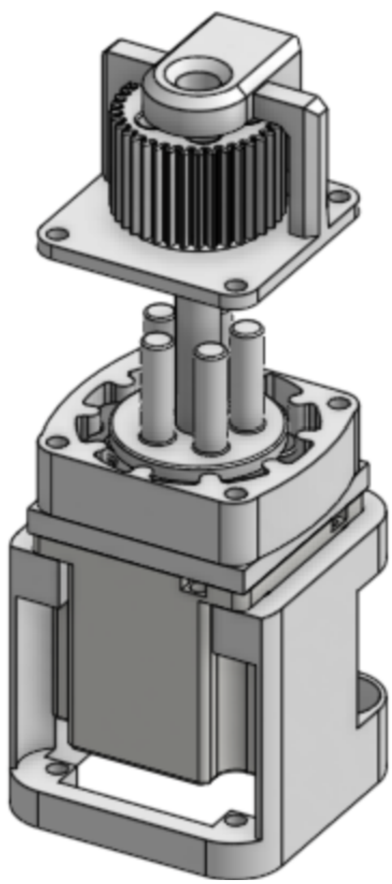


Figure 16: Exploded view of the carousel drive gear over the assembled gearbox.

14 The stack fully lowered and seated together.

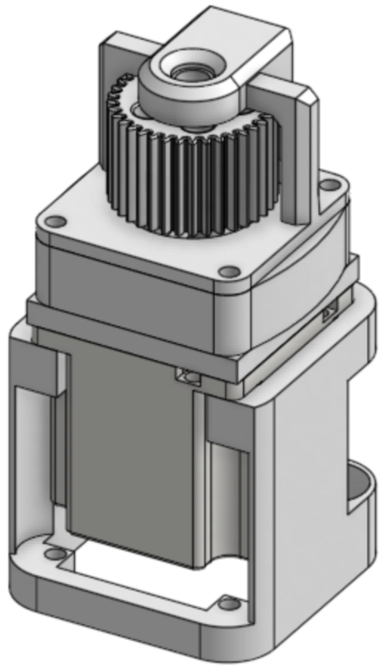


Figure 17: Carousel drive gear cover seated onto the completed gearbox assembly.

15 Insert four #10-32 1½" bolts into the holes and add matching nuts on the other side. Tighten by hand.

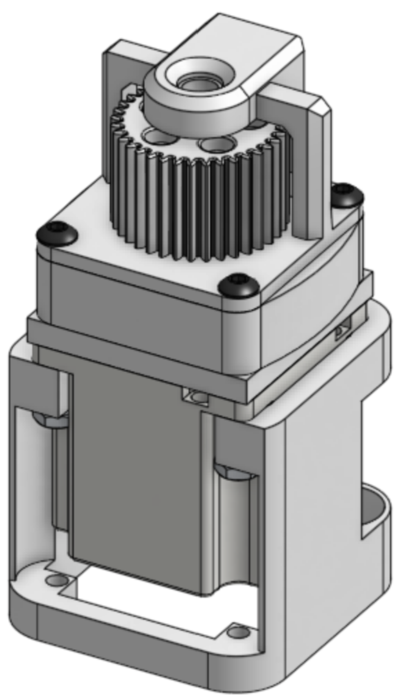


Figure 18: Four #10-32 bolts and nuts securing the gearbox cover stack.

Set the completed Carousel Motor and Gearbox assembly aside.

Part 3: Base Assembly

16 Lay out the four Base (BL, BR, FL, FR) components and the dovetail key that joins them.

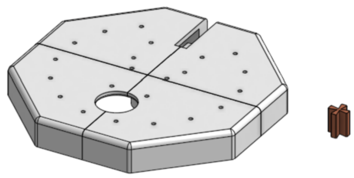


Figure 19: The four base quadrants and dovetail key, prior to assembly.

17 Apply glue to the dovetail, then insert it into the center junction of the four base components, ensuring it is properly aligned as shown below.

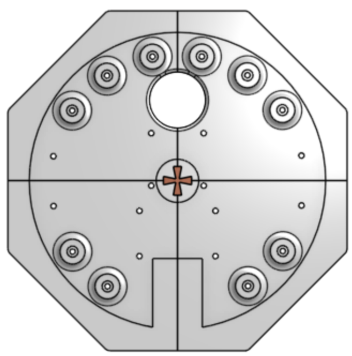


Figure 20: Dovetail key glued and inserted at the center junction of the four base quadrants.

18 Place the auger motor assembly (from **Part 1**) onto the base, aligning the mounting holes with the designated holes as shown below. Insert four #10-32 1" bolts from underneath the base and secure them with four matching nuts. Tighten the bolts to firmly fasten the auger motor assembly in place.

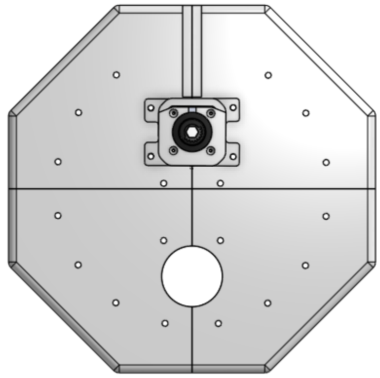


Figure 21: Auger motor assembly secured to the base plate using four #10-32 1" bolts and matching nuts.

19 A closer view confirming the motor mount's holes line up with the base before fastening.

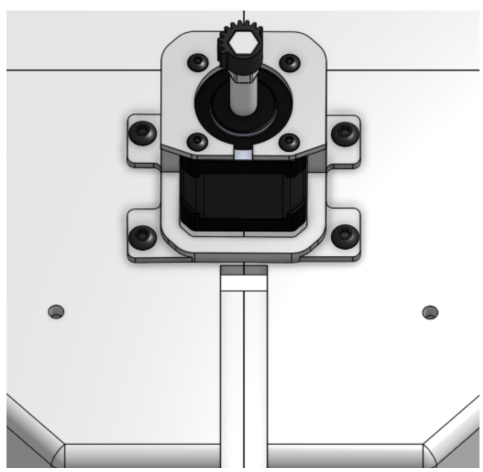


Figure 22: Close-up of the auger motor mount seated on the base.

20 Place the carousel motor and gearbox assembly (from **Part 2**) onto the base, as shown below. Insert four #10-32 1" bolts, this time with the bolt heads on the underside of the base and matching nuts on the topside.

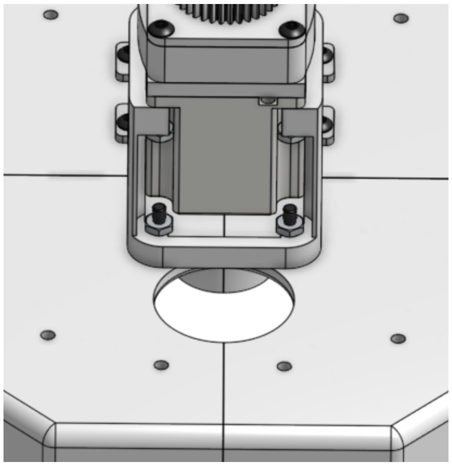


Figure 23: Carousel motor and gearbox assembly aligned above the base's center bore.

21 A view from underneath the base, showing the bolt heads seated against the base with the center dovetail key visible through the bore.

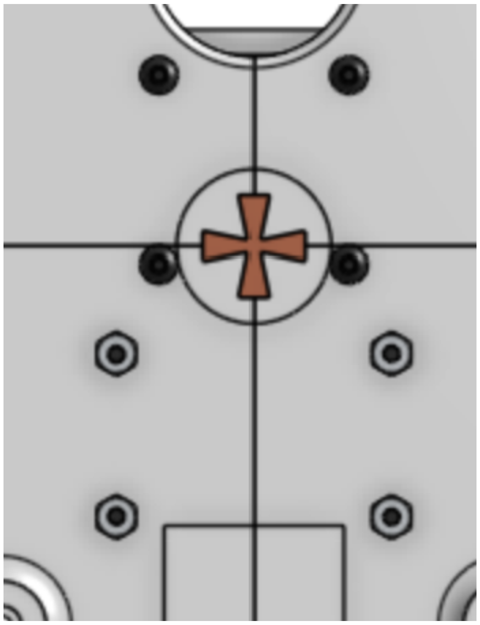


Figure 24: Underside view of the base showing bolt heads and the center dovetail key.

Set the Base assembly aside.

Part 5: Load Cell

22 With the load cell, super glue one of the load cell mounts to one of its ends, lining it up carefully.

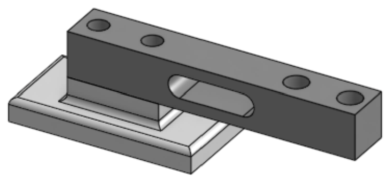


Figure 25: Load cell mount glued to one end of the load cell.

23 Super glue the load cell mount and base to the flat part on the inside of the base, as shown below.

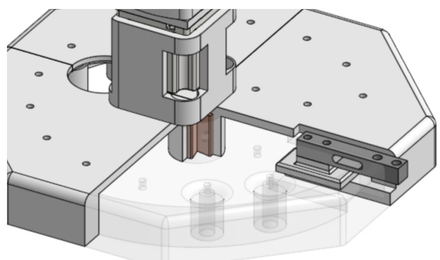


Figure 26: Load cell and mount glued into position on the inside of the base.

24 A closer view of the load cell mount seated flush against the base's inner wall.

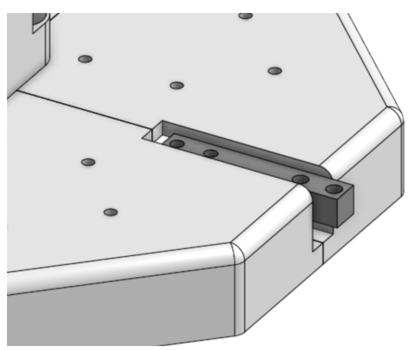


Figure 27: Close-up of the load cell mount glued to the base's inner face.

25 With the other load cell mount, super glue it to the opposite corner of the first load cell mount, facing upwards.

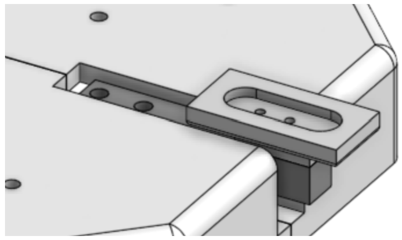


Figure 28: Second load cell mount glued facing upward at the opposite corner.

26 Then, super glue the Plate Holder onto the load cell mount.

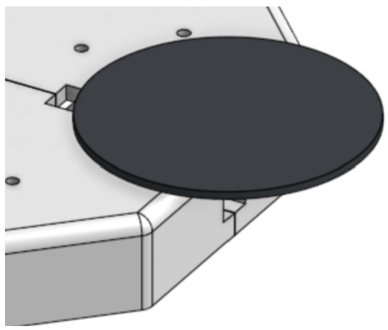


Figure 29: Plate Holder glued onto the upward-facing load cell mount.

Part 6: Carousel Segment

Build one segment completely, then repeat the whole part seven more times for a total of eight.

27 Place a bearing into the bearing well of one of the carousel segments, making sure it sits at the bottom of the well.

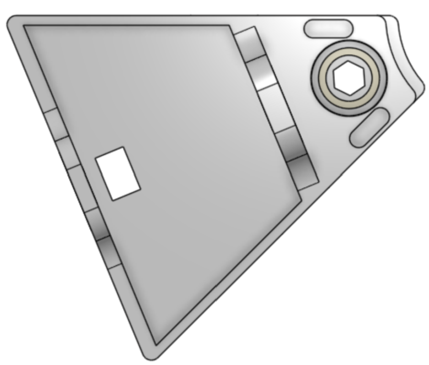


Figure 30: Bearing seated in a carousel segment's bearing well.

28 Then place a Segment (Container) into the slot. Set aside.

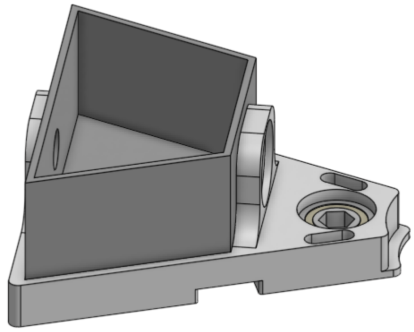


Figure 31: Segment container placed into the segment shell.

29 With the Segment (Insert), place an auger into the tunnel as shown below. Then carefully lower the insert/auger into the container.

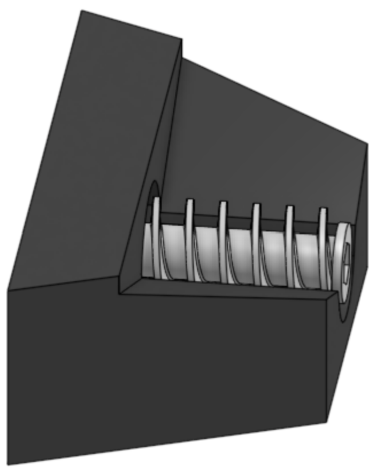


Figure 32: Auger placed into the tunnel of the segment insert.

30 The insert and auger lowered together into the container.

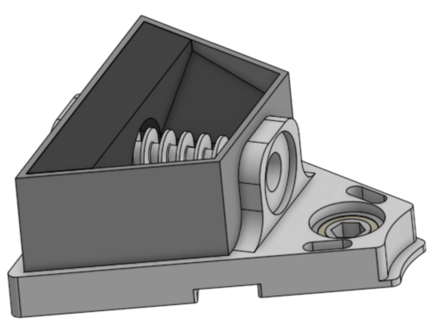


Figure 33: Segment insert and auger lowered into the segment container.

31 With the rear Auger End Cap, apply super glue to the hex side and insert it into the fitting in the auger. Repeat this process on the front end cap.

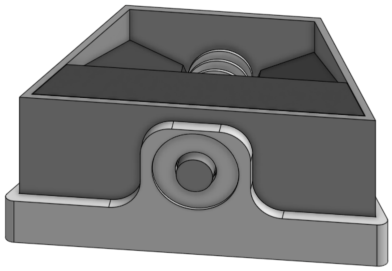


Figure 34: Rear auger end cap glued and inserted into the auger's hex fitting.

32 Front end cap and bearing detail, showing the hex bore that mates with the auger shaft.

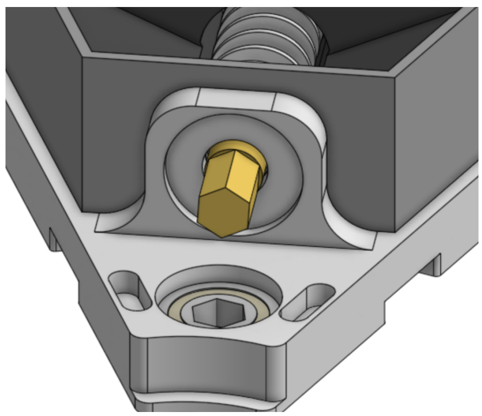


Figure 35: Front auger end cap and bearing hex-bore detail.

33 Turn the assembly over. With the Segment (Fork), apply super glue to the spaces shown below and insert the fork.

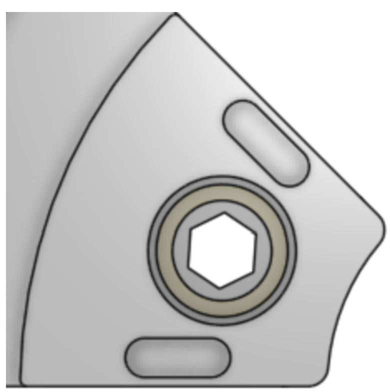


Figure 36: Segment fork glue points on the underside of the segment.

34 The fork seated and glued into place on the flipped segment.

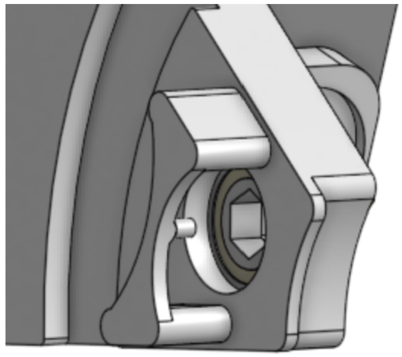


Figure 37: Segment fork inserted and glued in place.

35 Place one of the Auger Power Gears (Full Spur) onto the stud of the fork.

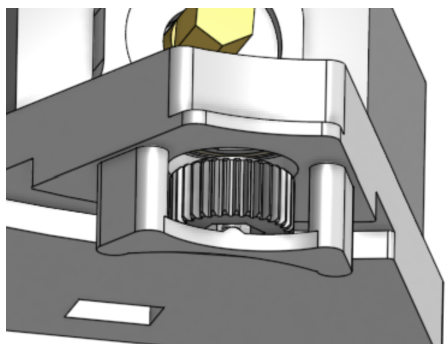


Figure 38: Full spur gear placed onto the fork's stud.

36 Place a Bevel Gear onto the auger end cap, applying a bit of super glue to the shaft beforehand.

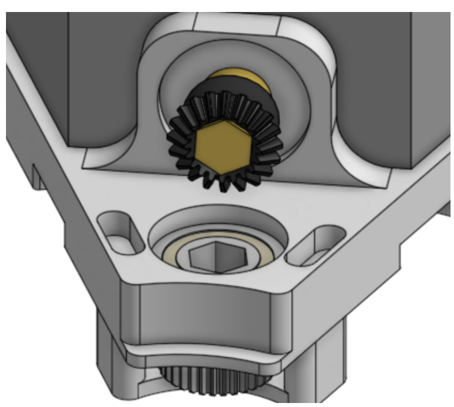


Figure 39: Bevel gear glued onto the auger end cap's shaft.

37 Place a matching bevel gear on the bearing, lining it up with the hex bore. Then insert the hex shaft through both gears.

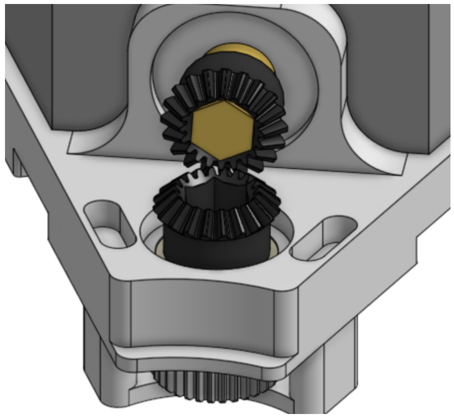


Figure 40: Matching bevel gear aligned to the bearing's hex bore.

38 The hex shaft fully seated, meshing both bevel gears.

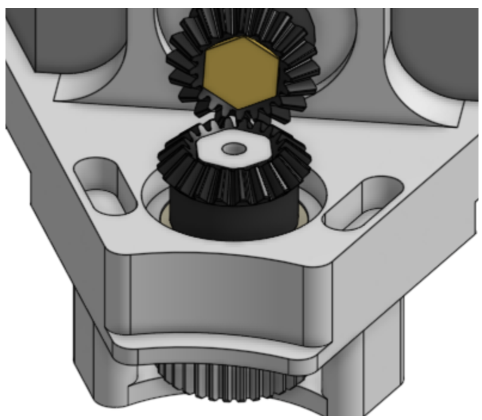


Figure 41: Hex shaft seated through the meshed bevel gear pair.

Repeat for All Eight

Repeat Steps 27–38 seven more times, for a total of eight completed segments, before continuing.

39 Line up all eight segments side by side, forming the full carousel ring.

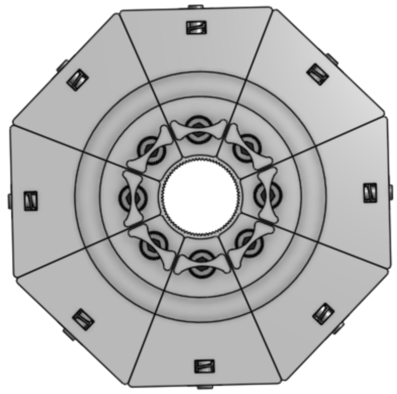


Figure 42: All eight segments arranged side by side, viewed from above.

40 The segments joined together into a single ring assembly.

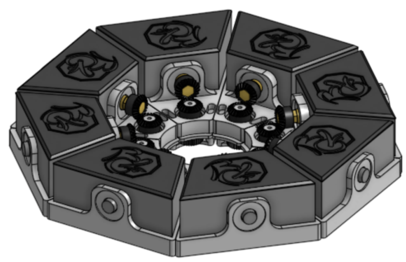


Figure 43: Eight segments joined into the assembled carousel ring.

41 With the ring gear, place it in the center of all the segments and super glue it in place.

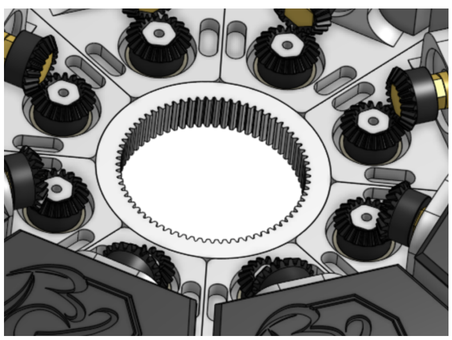


Figure 44: Ring gear glued into the center of the assembled carousel.

Set the completed carousel assembly aside.

Part 7: Tree Trunk

42 Place the lower trunk (identifiable as the one with the opening in the ring) onto the base plate, encompassing the gearbox assembly.

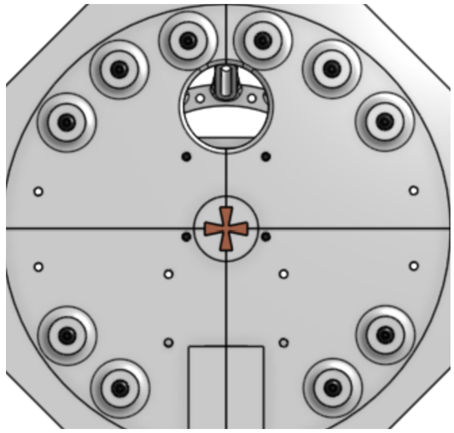


Figure 45: Lower trunk lowered onto the base plate around the gearbox assembly.

43 Insert a #10-32 $\times$ $\frac{3}{4}$" bolt through each mounting hole in the base, then secure it with a matching nut on the topside. Ensure that all bolts are properly seated and tightened.

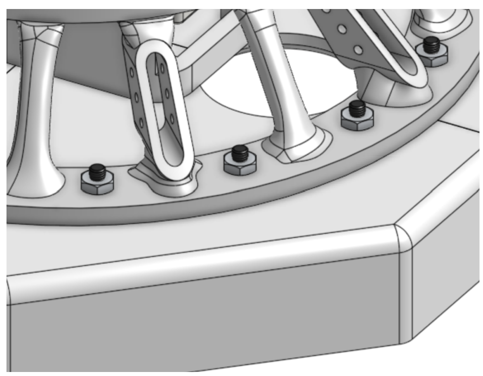


Figure 46: Support arms secured to the base plate using #10-32 bolts and nuts.

44 The lower trunk fully seated on the base, with the carousel drive and motor visible through the center opening.

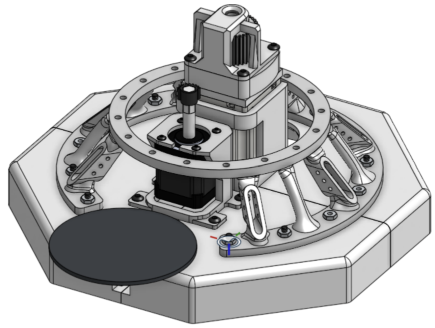


Figure 47: Lower trunk seated on the base, carousel drive visible through the center.

45 Place the upper trunk onto the lower trunk, lining up the holes between the two. With the support plate, place it as shown.

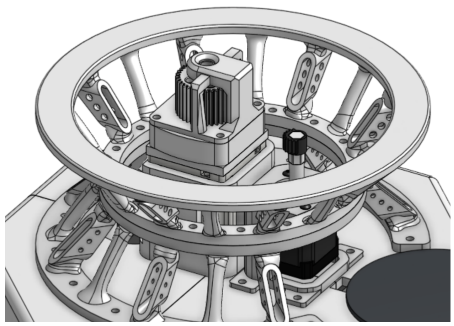


Figure 48: Upper trunk aligned onto the lower trunk with the support plate positioned.

46 The upper and lower trunk sections joined, ready for fastening.

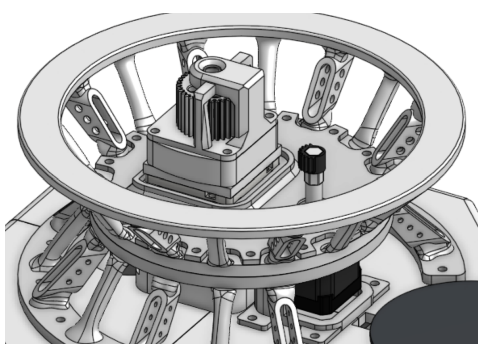


Figure 49: Upper and lower trunk sections joined together.

47 Insert a #10-32 $\frac{3}{4}$″ bolt and matching nut through each of the mounting holes as shown. Ensure that the bolts are properly seated and tightened.

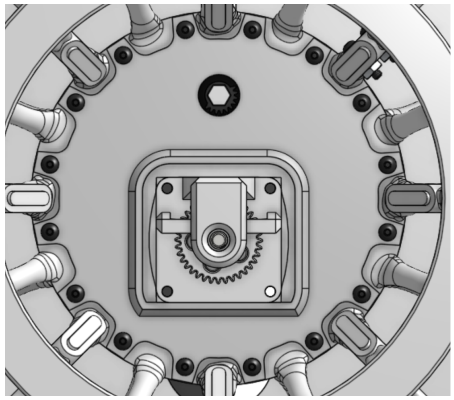


Figure 50: #10-32 $\frac{3}{4}$″ bolts and matching nuts installed through the trunk's aligned mounting holes.

48 Following the image below as a guide, insert the support arms and funnel into the correct slots, then secure each component using two #10-32 $\frac{3}{4}$″ bolts and matching nuts.

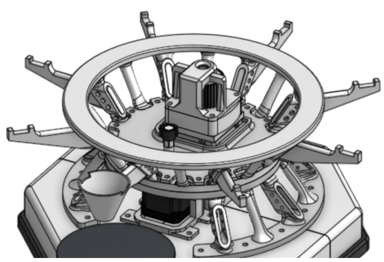


Figure 51: Support arms and funnel secured in the trunk's slots using #10-32 $\frac{3}{4}$″ bolts and matching nuts.

49 Support arms and funnel fastened in place with bolts and nuts.

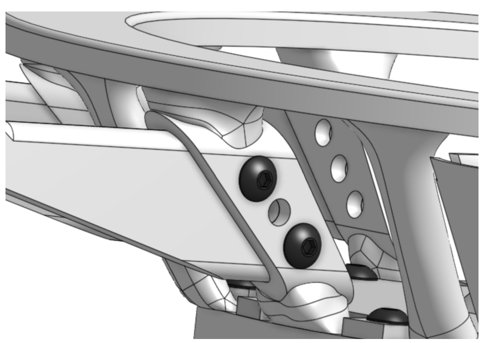


Figure 52: Support arms and funnel secured with bolts and nuts.

Part 8: Final Assembly

50 With the support ring, place it on the support arms so the opening sits above the funnel. Add super glue at the seams to lock it in place.

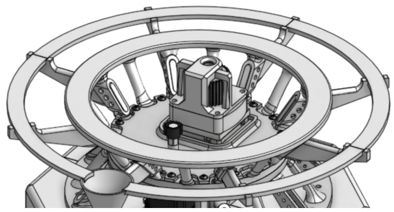


Figure 53: Support ring placed on the support arms above the funnel.

51 With the assembled carousel (from **Part 6**), place it on the ring track of the upper trunk, making sure the ring gear meshes with the carousel drive.

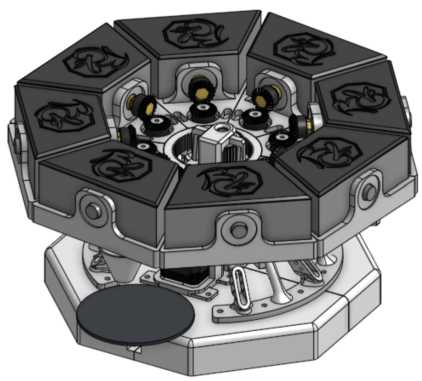


Figure 54: Assembled carousel lowered onto the upper trunk's ring track.

52 Place all the electronics and wiring into the base (**Section ??**), then enclose it with the base plates.

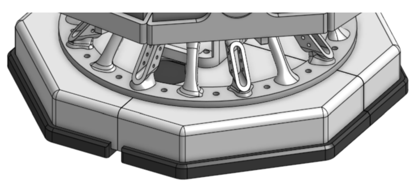


Figure 55: Electronics and wiring routed into the base before closing with the base plates.

ⓘ Mechanical Build Complete

With **Part 8** finished, the mechanical assembly is done. Continue to **Section ??** for the electrical wiring if it isn't already in place, then to **Section ??** to install and launch the software.

2.3 Electrical Wiring

All wiring is routed beneath the base plate, keeping the carousel deck and inside the tree trunk clear of cable clutter. The PERF board, stepper drivers, and HX711 are mounted to the underside of the base; only the motor leads pass up through the base into the tower.

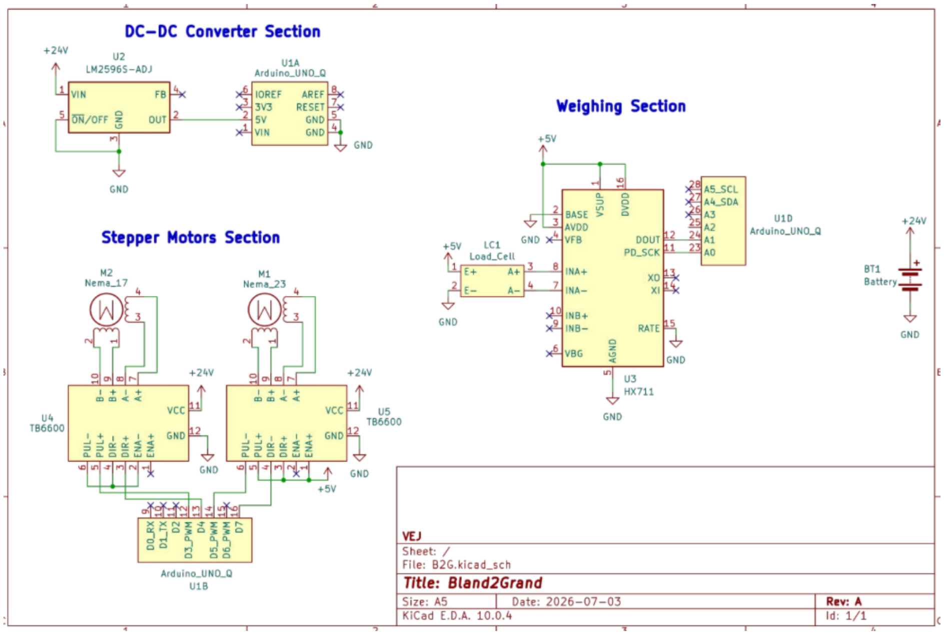


Figure 56: Full electrical schematic for Bland2Grand.

1. Plug the 4-pin wires into the NEMA 23 and NEMA 17 motors.

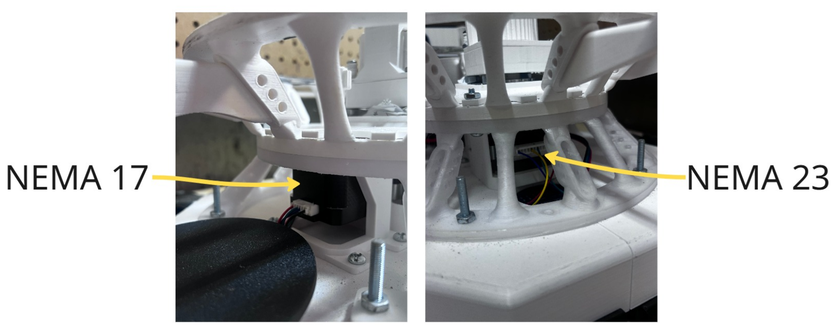


Figure 57: 4-pin wire connectors seated into the NEMA 23 and NEMA 17 motors.

2. Thread the wires through the routing hole shown below, down into the base.

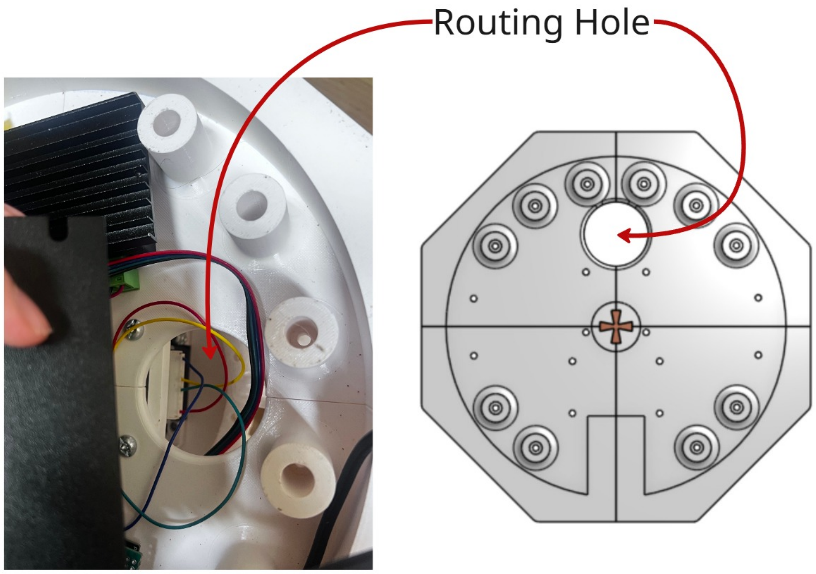


Figure 58: Motor wires threaded through the routing hole into the base.

3. Wire and solder the remaining connections according to the schematic above.

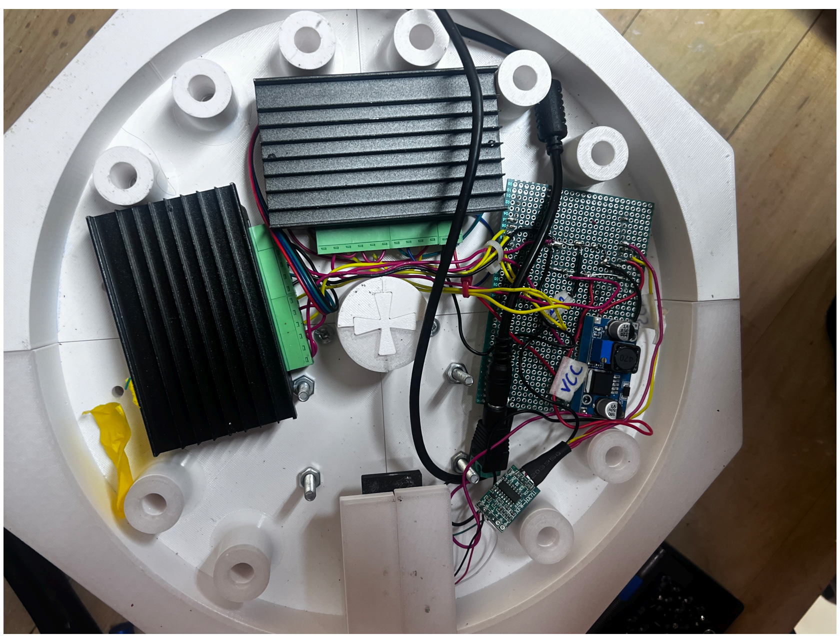


Figure 59: Finished wiring, soldered and connected per the schematic.

What's Underneath the Base

- **Controller board** (UNO Q or UNO R4 WiFi): centered on the sub-plate, standoff-mounted for airflow and easy access to the USB port.

- **Stepper drivers**: one per auger motor, mounted in a row along one edge, each wired to its corresponding controller pin group per the schematic.
- **HX711 load cell amplifier**: mounted close to the load cell itself to keep that signal run short, then a single 4-wire cable carries the amplified signal up to the controller.
- **Power distribution terminal**: splits the 24 V input to the motor drivers and steps it down as needed for the controller and load cell amplifier.

> **! Disconnect power before wiring changes**
>
> Always unplug the 24 V supply before connecting or disconnecting anything from the base. The stepper drivers in particular can be damaged if motor leads are connected or removed while powered.

3 Code and Software Setup

All source code for Bland2Grand is hosted on GitHub: `github.com/ElliottStarosta/Bland2Grand`. The repository contains separate, self-contained builds for each supported board. Set up the one that matches your hardware.

3.1 Arduino UNO Q Setup (Recommended)

The UNO Q build runs the whole system (MCU sketch, Flask backend, REST API, SSE stream, and the built React frontend) as a single App Lab app on the board itself. There is no separate laptop backend/frontend to run.

Phase 1: Download the Project

1. Navigate to the Bland2Grand GitHub repository.

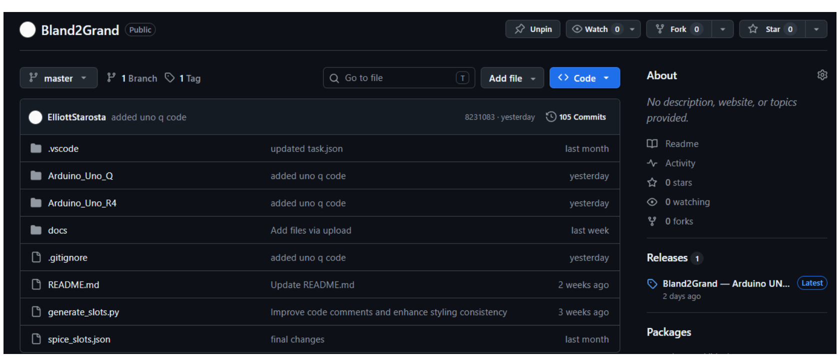


Figure 60: The Bland2Grand GitHub repository page.

2. Locate and click on the latest release.

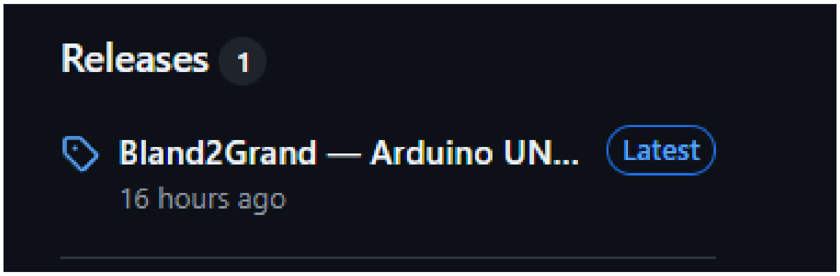


Figure 61: Latest release on the Releases page.

3. Scroll to the **Assets** section at the bottom of the release.

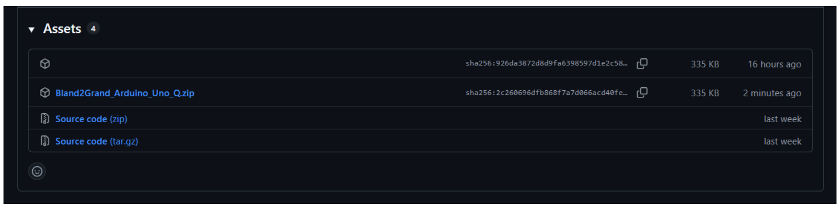


Figure 62: Assets section at the bottom of a release.

4. Download the zip file: `Bland2Grand_Arduino_Uno_Q.zip`.

Figure 63: Downloading `Bland2Grand_Arduino_Uno_Q.zip` from the Assets list.

Phase 2: Deploy to Arduino / App Lab

5. Open **App Lab**.

6. Select your board.

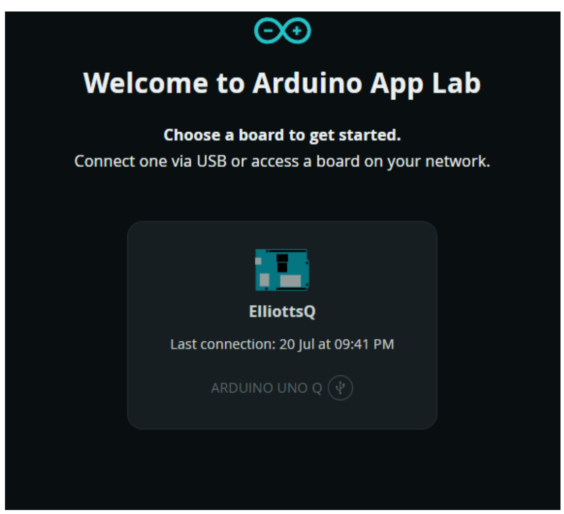


Figure 64: Selecting the connected UNO Q board in App Lab.

7. Click **Create new app** in the top right-hand corner.

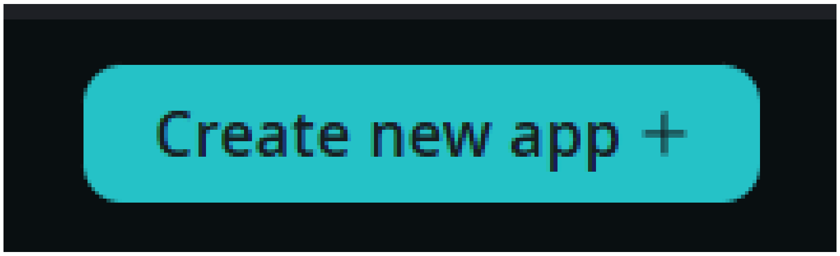


Figure 65: The "Create new app" button in App Lab.

8. Click **Import App** and select the `.zip` file you downloaded in Phase 1.

Phase 3: Configure API Credentials

The AI-assisted custom blend feature calls out to OpenRouter. An API key is required for this feature to work (the rest of the app — search, dispensing, calibration — works without it).

9. Go to the OpenRouter keys page.

10. Click **New Key**.

Figure 66: The "New Key" button on the OpenRouter keys page.

11. Fill out the required information.

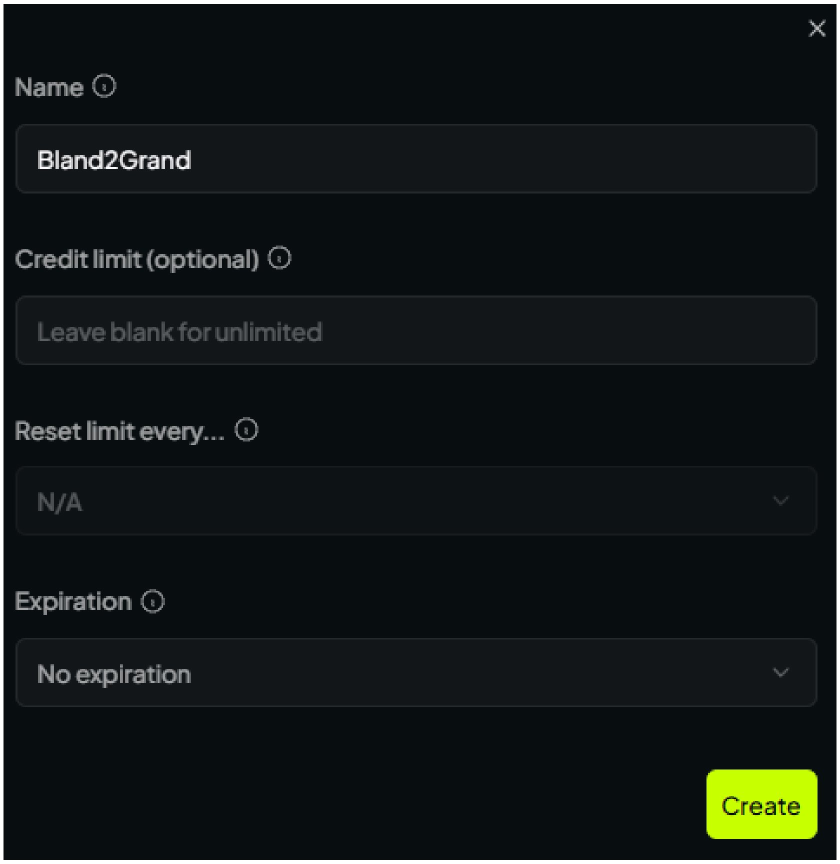


Figure 67: OpenRouter new key creation form.

12. Copy the key shown on the confirmation screen right away — OpenRouter will not show it to you again.

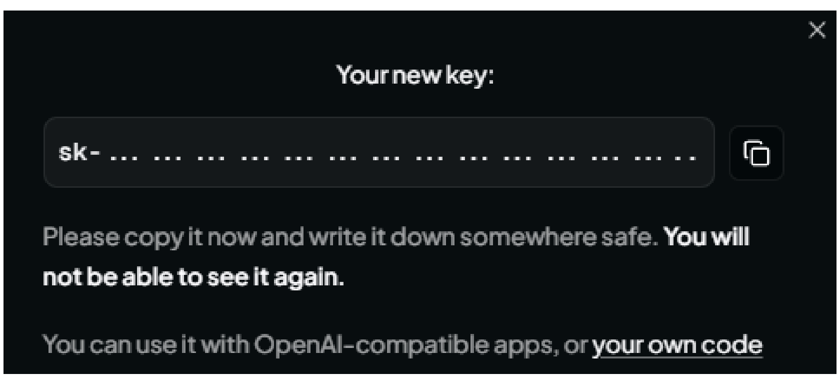


Figure 68: The one-time key reveal screen after creating a key.

13 Update the relevant environment variables for the app, most importantly `OPENROUTER_API_KEY`.

```
OPENROUTER_API_KEY=YOUR_KEY_HERE
AI_MODEL=nvidia/nemotron-3-super-120b-a12b:free
DATABASE_PATH=/app/python/bland2grand.db
MOCK_BRIDGE=false
FLASK_PORT=5000
FRONTEND_DIST=./dist
```

Figure 69: Example environment configuration for the UNO Q app.

> **Keep your API key private**
>
> Treat your OpenRouter key like a password: don't commit it to a public repository, share it in screenshots, or paste it into chat tools. If a key is ever exposed, revoke it from the OpenRouter dashboard and generate a new one.

Phase 4: Run the Application

13 Set the application to run on startup by clicking the down arrow at the top left-hand side of the screen, next to "Bland2Grand".

Figure 70: The startup dropdown next to the app name in App Lab.

14 Toggle the **Run at startup** option to enabled.

Figure 71: Enabling the "Run at startup" toggle.

15 Click the **Run** button in the top right-hand corner of the screen. App Lab uploads the MCU sketch and starts the Flask app on the board's Linux side.

16 Access the web server by opening the board's address on port **5000** (the port declared in `app.yaml`) in your browser.

17 You are now ready to use Bland2Grand.

Why enable Run at startup

With **Run at startup** enabled, the UNO Q automatically relaunches the Bland2Grand app any time it's powered on or rebooted, so you don't need App Lab open to bring the machine back up after a power cycle.

No separate frontend server

Unlike the UNO R4 build below, the UNO Q serves the built React frontend directly from the same Flask app that handles the API, so there is nothing else to start or forward — one board, one port, one app.

3.2 Arduino UNO R4 WiFi Setup

The UNO R4 build splits the system across three pieces: firmware on the Arduino, a Flask backend, and a React frontend, both of the latter run on a laptop and communicate with the Arduino over WiFi. This section assumes the hardware is already assembled and wired per **Section 2**, and that the `Bland2Grand_Arduino`, `bland2grand-backend`, and `bland2grand-frontend` folders (and the provided `.vscode/tasks.json`) are on the laptop that will run the software.

Hardware Power-On

1. Place the machine on a flat, level surface — the load cell platform must stay level to within 2° for accurate readings.
2. Connect the 24 V power supply to the wall.
3. Confirm each of the 8 spice containers is loaded and seated.
4. Place an empty bowl on the scale platform. The scale tares before each dispense, so the exact bowl weight does not matter.
5. Make sure the host laptop is powered on and ready to start its mobile hotspot (see below). The Arduino and the phone running the web app both need to join it.

Launching the Software

All software is launched from VS Code using the provided `tasks.json`. Open the project workspace in VS Code, then open the Command Palette with **Ctrl+Shift+P** and select **Tasks: Run Task**.

Figure 72: VS Code Command Palette showing "Tasks: Run Task" selected.

Two compound tasks are available:

Task Name	What it does
Bland2Grand: Full Stack	Flashes firmware to the Arduino, opens the serial monitor, starts the Flask backend, and starts the Vite frontend, which is all in parallel. Use this for a full bring-up from scratch.
Bland2Grand: Backend + Frontend	Starts only the Flask backend and the Vite frontend. Use this when the Arduino is already flashed and running. This is the task to run for a demo.

ⓘ What Full Stack does automatically

The **Arduino: Flash + Serial** subtask does three things before flashing: it starts the Windows mobile hotspot (so the Arduino has a network to connect to), kills any open serial monitor that would block the port, then runs `pio run --target upload` followed by `pio device monitor --baud 9600`. No manual hotspot setup or port management is needed.

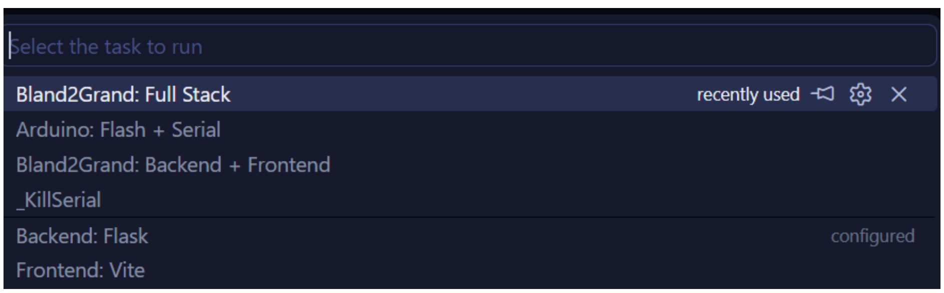


Figure 73: VS Code task picker showing the available Bland2Grand tasks.

Once the **Backend + Frontend** task is running, the web app is accessible from any device on the same network at:

`http://{laptop_ip}:5173`

The laptop's IP address is printed in the Vite terminal panel when it starts. On the Arduino's serial monitor you will also see the IP it received from the hotspot, which confirms the two devices are on the same subnet.

First-Time Backend Setup

If this is the first time running the backend on this laptop, set it up once before using the VS Code tasks:

```
cd bland2grand-backend
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
python seed_recipes.py
```

Create a `.env` file in `bland2grand-backend/`:

```
ARDUINO_URL=http://192.168.x.x
OPENROUTER_API_KEY=YOUR_KEY_HERE
AI_MODEL=anthropic/claude-3-haiku
DATABASE_PATH=bland2grand.db
FLASK_PORT=5000
MOCK_ARDUINO=true
```

```
cd bland2grand-frontend
npm install
```

Open `Bland2Grand_Arduino` in PlatformIO once to confirm it builds before relying on the VS Code task to flash it.

Set `MOCK_ARDUINO=false` for a real run

`MOCK_ARDUINO=true` simulates dispensing in software with no hardware attached. This is useful for testing the app without the machine. Set it to `false` once the Arduino is flashed and wired, so the backend talks to the real hardware.

4 Using Bland2Grand

Once your board is flashed and the app is running (**Section ??** or **Section ??**), everything from here on is identical regardless of which board you built. You interact with Bland2Grand entirely through a web app on your phone: the machine handles all indexing, dispensing, and weight confirmation autonomously and reports the result back in real time.

Figure 74: Bland2Grand assembled system, ready to dispense.

4.1 Powering On and Preparing to Dispense

1. **Place the machine on a flat, level surface.** The load cell platform must stay level to within 2° for accurate readings — avoid uneven countertops or surfaces that vibrate.

2. **Connect power to the board.** 24 V for the UNO R4 build, or the UNO Q's own power input if that's your build.

3. **Verify spice containers are loaded.** Each of the 8 slots should hold its designated spice (see the slot map in **Section ??**). Friction-fit lids pull straight off — do not twist.

4. **Place an empty bowl on the scale platform.** Sit it flat and centred — the scale tares before each dispense, so the exact bowl weight doesn't matter.

Figure 75: Bowl placed on the scale platform at the base of the machine.

5. **Confirm the app is reachable.** Open the board's URL on your phone (port **5000** for UNO Q, or `http://{laptop_ip}:5173` for UNO R4) and make sure the phone is on the same network as the board.

> **Same app, either board**
>
> The interface below is served identically whether the Flask app is running on the UNO Q itself or on a laptop for the UNO R4 build. Once the URL loads, the rest of this section applies exactly as written.

4.2 Using the Mobile Web App

Idle Screen: Tap to Wake

When you first open the URL, the idle screen appears: ambient coloured orbs drift across a dark background with the Bland2Grand logo and a pulsing tap indicator. Tap anywhere to wake the app.

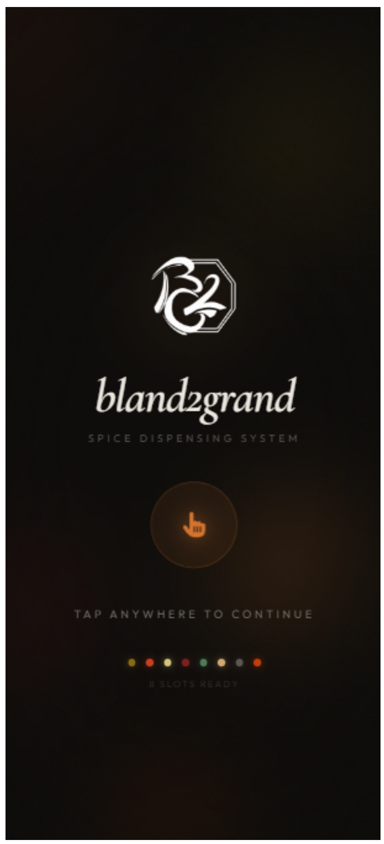


Figure 76: Idle screen. Tap anywhere to continue.

Search Screen: Find a Recipe

The search screen offers three ways to find a recipe:

1. **Type a dish name** into the search bar. Results appear automatically after a 2.5-second pause. If no local match exists, the AI generates a blend and saves it to the database for next time.

2. **Tap a Featured Blend** tile to jump directly to a popular recipe without searching.

3. **Tap a cuisine category** (Mexican, Indian, Italian, etc.) to browse everything in that category.

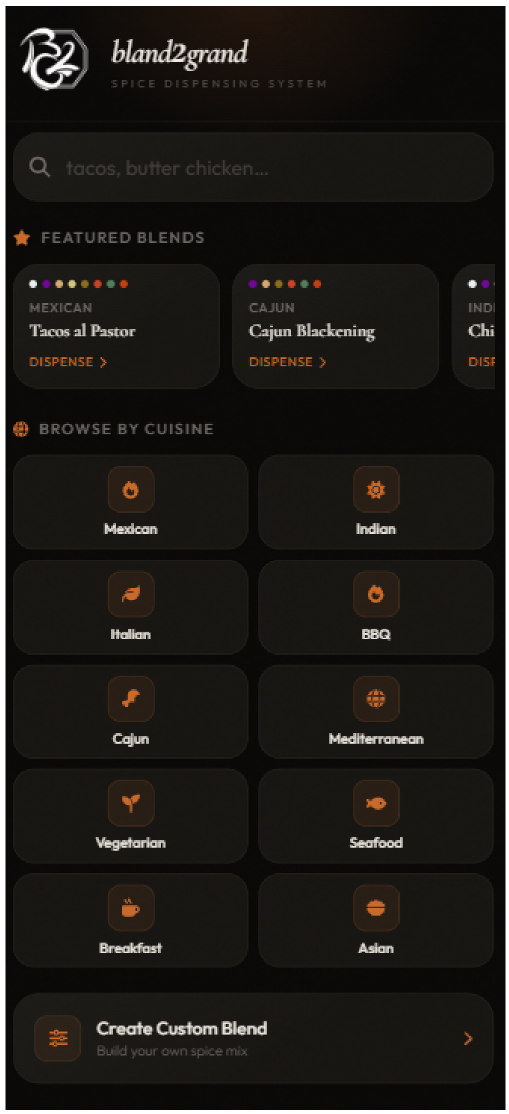
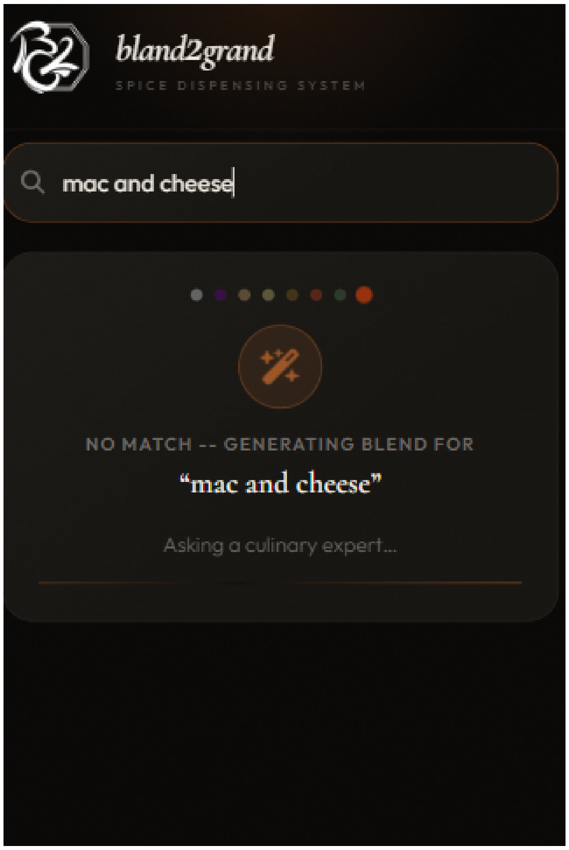


Figure 77: Search screen showing the search bar, featured blends, and cuisine category grid.

If the dish isn't in the local database yet, an AI loading overlay appears while the blend is generated; this takes 1–3 seconds and only happens once per unique dish name. Every future search for the same dish is served instantly from the cache.



AI generating a blend

Generated result

Figure 78: AI loading overlay shown while generating a new recipe, and the resulting spice blend.

- **Results Screen: Choose Your Blend**

Up to six matching recipes appear as cards, each showing the recipe name, a cuisine badge, a proportional spice colour bar, and the gram amount per serving for every active spice. Tap a card to select it.

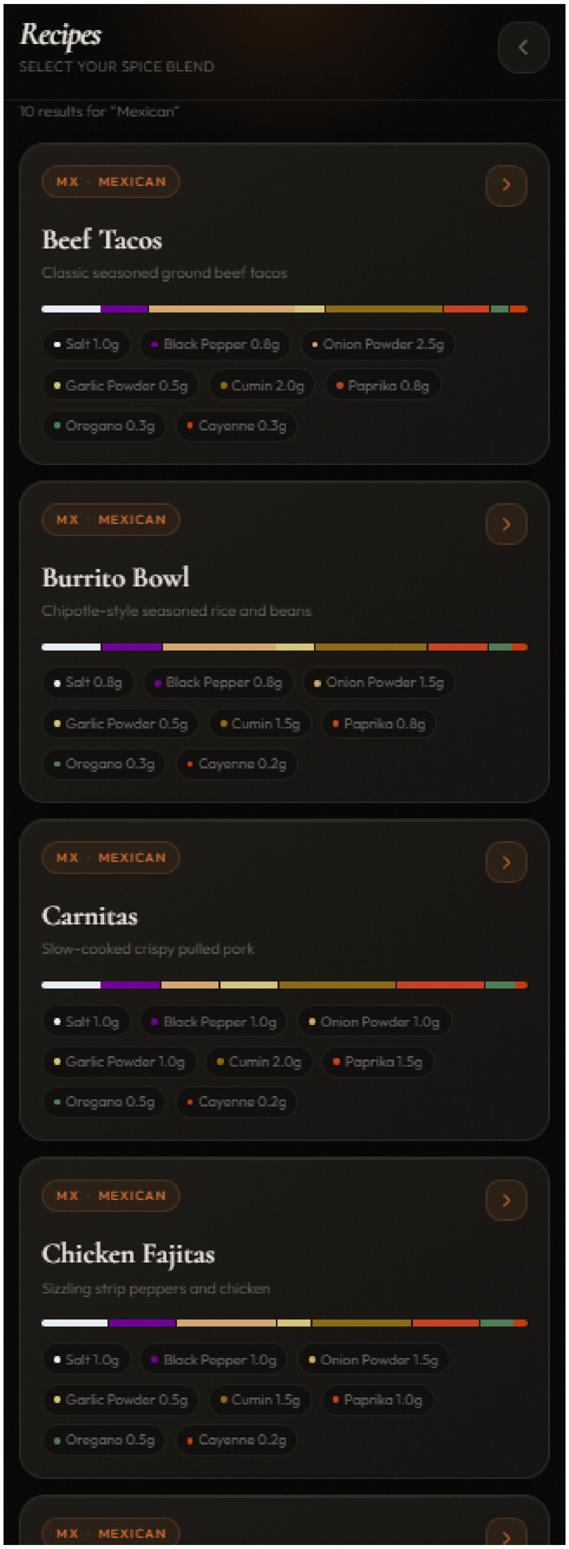


Figure 79: Results screen showing matching recipe cards with spice colour bars and gram breakdowns.

Serving Screen: Set Portions

Use the large +/- buttons or the quick-pick row (1, 2, 4, 6, 8) to set how many servings to dispense. Gram amounts update live as you adjust. When ready, tap **Dispense N servings**.

Figure 80: Serving screen with the counter set and gram breakdown updating live.

Dispensing Screen: Watch It Work

While dispensing, the screen shows:

- A **bowl visualisation** that fills with layered spice in each slot's colour as weight accumulates.
- A **per-spice progress row** with a live bar, status icon (rotating for indexing, spinning for dispensing, checkmark for done), and current vs. target weight.
- A **circular progress ring** showing overall completion percentage.
- A **status line** showing the current action (e.g. *"Rotating…"* or *"Cumin"*).

Each spice plays a voice announcement when the carousel reaches its slot.

Don't remove the bowl mid-dispense

Pulling the bowl off the scale during dispensing will throw off the weight readings for the rest of the session. Wait for the Complete screen before touching it.

Figure 81: Dispensing screen mid-session: bowl filling, per-spice rows, and progress ring.

Complete Screen: Your Blend Is Ready

Once every spice has been dispensed, the screen transitions to Ready, showing:

- A green checkmark (or amber warning if any slot timed out).
- A summary card listing the actual dispensed weight for each spice against its target.
- The total dispensed weight.

Retrieve your bowl and start cooking. Tap **Start a new blend** to return to the search screen.

Figure 82: Complete screen showing the dispensed summary with accuracy indicators.

Custom Recipe Builder

Tap **Create Custom Blend** on the search screen to build your own blend from scratch:

1. Enter a blend name and optional description at the top.
2. For each of the 8 slots, tap the unit label (`tsp`, `tbsp`, or `g`) to cycle between units. The amounts convert automatically using each spice's density constant.

3. Use the +/- buttons to set the amount for each spice. A live proportional bar chart updates as you adjust.

4. Tap **Save & Dispense** to write the recipe to the database and proceed to the serving screen.

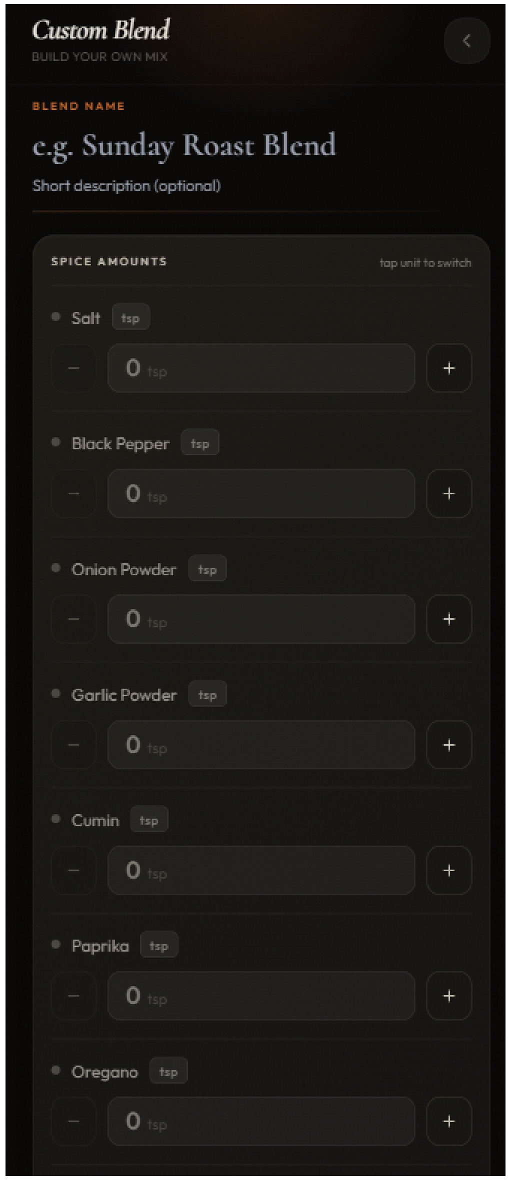


Figure 83: Custom recipe builder showing per-slot amount controls, unit picker bubbles, and live spice bar.

4.3 Refilling Containers

Each container has a friction-fit lid. Ensure the machine is not actively dispensing before opening the lid. Fill the container, then replace the lid, pressing it down until it is firmly seated.

Slot	Spice	Slot	Spice
1	Salt	5	Cumin
2	Black Pepper	6	Paprika
3	Onion Powder	7	Oregano
4	Garlic Powder	8	Cayenne

Don't swap spices between slots without recalibrating

Each slot's calibration factor is measured for that specific spice's bulk density. Swapping spices without running a new calibration will produce inaccurate dispense amounts. Use the `POST /api/calibrate` endpoint, or the calibration sketch in `src/tests/calibration.cpp`, to update a slot's factor after any spice change.

4.4 Safety Notes

- A **60-second per-spice timeout** halts the auger and aborts the blend if the target weight isn't reached; protects against empty containers or a jammed auger.
- A **30-second watchdog** halts all motors if the connection to the backend is lost mid-session.
- A **scale overload fault** triggers if more than 1 500 g is detected on the platform.
- Do not reach into the carousel area while the machine is powered on.